Answers- stacks and queues

1. ABCD+*F/+DE*+
2. 142

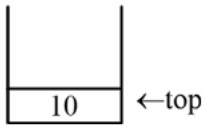
   Using stack: • Scan ⇒ left to right (i) Operand ⇒ Push it (ii) Operator

   (a) Pop top 2 elements from stack let first popped element is A & second popped elements is B.

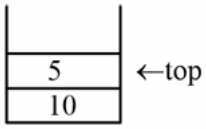
   (b) Evaluate B +A (where + is the operator found)

   (c) Push the result of previous step onto stack. Keep repeating the above, once the input is over at the end, stack top element is the answer. 10, 5, +, 60, 6, /, *, 8, –

   Push 10:

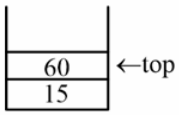
   

   Push 5:

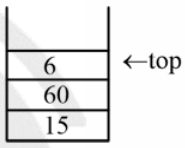
   

   + Operator is coming, pop the top two elements and calculating the result. Result = 10 + 5 = 15 And push the result into stack.

   Push 60:

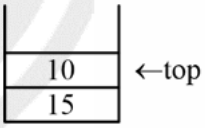
   

   Push 6:

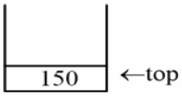
   / Operator is coming, pop the top two elements and calculating the result. Result = 60 / 6 = 10. And push the result into stack.

   Push 10 (result):

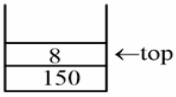
   

   * Operator is coming, pop the top two elements and calculating the result. Result = 15 × 10 = 150 And push the result into stack.

   Push 150 (result):

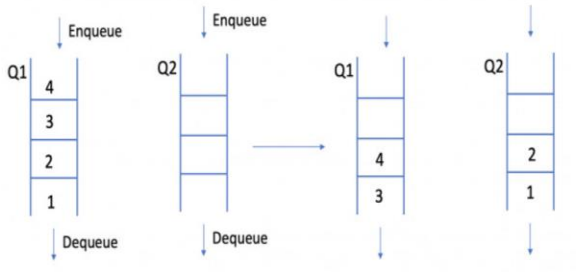
   

   Push 8:

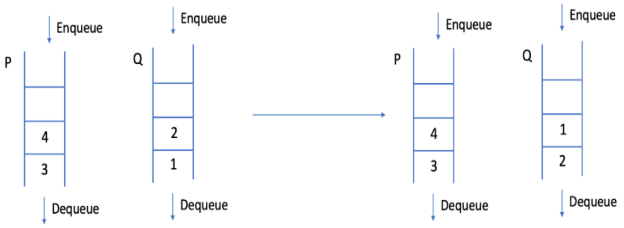
   

– Operator is coming, pop the top two elements and calculating the result. Result = 150 – 8 = 142

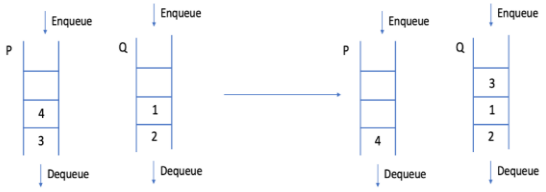
3. a3,8
4. 0 Enqueue to Q1
   Step 1:



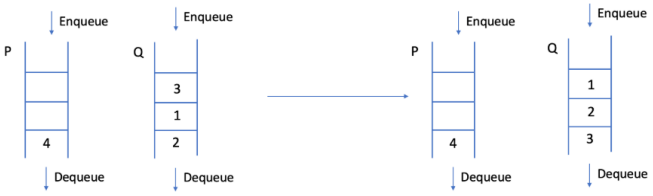
Step 2: Now reverse Q2. This doesn't cost any ENQUEUE to Q1



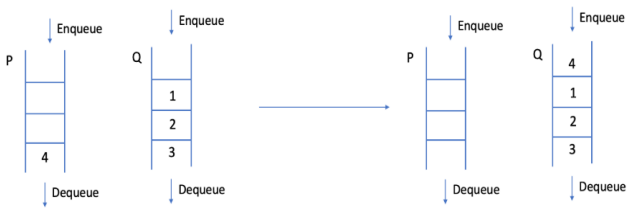
Step 3:



Step 4: Reverse Q2



Step 5:



Step 6:



5. Full: (REAR + 1) mod n = = FRONT ;Empty: REAR = = FRONT
6. 256
   The worst case happens when the queue is sorted in decreasing order. In worst case, loop runs n*n times.
   Lets take queue have 4 element then
   Queue: 4 3 2 1
   Stack: Empty

   3 2 1
   4

   3 2 1 4
   Empty

   2 1 4
   3

   2 1 4 3
   Empty

   1 4 3
   2

   1 4 3 2
   Empty

   4 3 2
   1

   3 2
   1 4

   3 2 4
   1

   2 4
   1 3

   2 4 3
   1

4 3
1 2

3
1 2 4

3 4
1 2

4
1 2 3

Empty
1 2 3 4
So Total iteration=16(4X4)
So if queue have 16 element then no of iteration=256(16X16).

7. D
8. 17

Element	Push ends at (s)	Pop starts at (s)	Life (s)
5	19	20	1
4	15	24	9
3	11	28	17
2	7	32	25
1	3	36	33

Sum of lives=1+9+17+25+33=85
Avg stack Life=85/5=17

9. ++-+KL*MN*//*^OPWUVTQ
10. b) 3, 4, 5, 2, 1

option A:
Push 1, Push 2, Push 3, Pop → 3
Push 4, Pop → 4
Push 5, Pop → 5
Stack now: [1, 2] (2 on top)
Need next output = 1, but top of stack = 2 → Invalid
Option B:
Push 1, Push 2, Push 3, Pop → 3
Push 4, Pop → 4
Push 5, Pop → 5
Stack now: [1, 2] (2 on top)
Pop → 2
Pop → 1
Valid
Option C:
Push 1, Pop → 1
Push 2, Push 3, Push 4, Push 5, Pop → 5
Stack now: [2, 3, 4] (4 on top)
Need next output = 2, but top = 4 → Invalid
Option D:
Push 1, 2, 3, 4, 5
Pop → 5, Pop → 4, Pop → 3
Stack now: [1, 2] (2 on top)

Need next output = 1, but top = 2 → Invalid